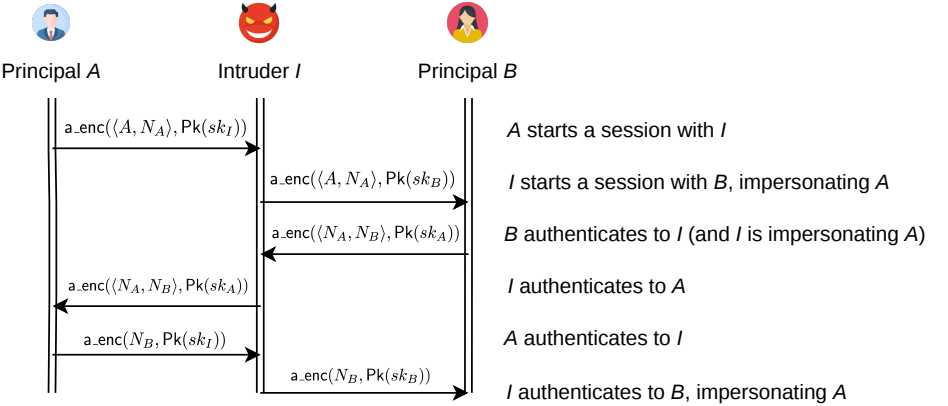
Principal A
Intruder I
Principal B
$\mathsf{a_enc}(\langle A, N_A\rangle, \mathsf{Pk}(sk_I))$
A starts a session with I
$\mathsf{a_enc}(\langle A, N_A\rangle, \mathsf{Pk}(sk_B))$
I starts a session with B, impersonating A
$\mathsf{a_enc}(\langle N_A, N_B\rangle, \mathsf{Pk}(sk_A))$
B authenticates to I (and I is impersonating A)
$\mathsf{a_enc}(\langle N_A, N_B\rangle, \mathsf{Pk}(sk_A))$
I authenticates to A
$\mathsf{a_enc}(N_B, \mathsf{Pk}(sk_I))$
A authenticates to I
$\mathsf{a_enc}(N_B, \mathsf{Pk}(sk_B))$
I authenticates to B, impersonating A